

The Model in Three Pages

Multinational ownership, market power, and the measurement of foreign profits

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1 Introduction

A large share of Latin American exports is shipped by subsidiaries of international groups headquartered elsewhere — often in the very country imposing the tariff. **How much of the profit earned in those markets belongs to that country?** The obvious answer — total profit times the share of firms owned — is wrong, and Proposition 1 gives the exact error. Getting there needs three things at once: markups that rise with firm size, multinational production, and entry. Each is standard; the combination, plus ownership recorded *firm by firm*, is not. Section 2 sets out the model, Section 3 derives the measurement result, Section 4 states what is proved about uniqueness, and Section 5 confronts the six facts.

2 Model

2.1 Set-Up

Agents and markets. There are N countries and K sectors; a *market* is one sector in one destination country. Three kinds of actors: a representative *household* in each country, which supplies labour L_n inelastically and spends its income across markets (Cobb–Douglas weights β_k across sectors, CES within a market with elasticity σ ; substitution across markets is η , with $\sigma > \eta \geq 1$); *ownership groups* g — multinational parents with a potential factory in every country, and single-factory local firms — each headquartered somewhere and owned by countries in shares θ_{gn} ; and the *factories* they operate, which produce differentiated varieties and sell them across borders. The pool of potential parents grows with country size, while the capability distribution they draw from is identical everywhere — so rich countries host more parents only because more draws have a better maximum.

From: The two-tier demand is Atkeson and Burstein (2008); the granular pool of parents is Gaubert and Itskhoki (2021).

Technology and delivered costs. Factory a (head office h , located in ℓ , selling to n , sector k) delivers at unit cost

$$c_{a,n} = \frac{1}{\varphi_a} \underbrace{w_\ell^{1-\nu_k}}_{\text{wages}} \underbrace{(P_{\ell k}^{\text{IO}})^{\nu_k}}_{\text{inputs}} \underbrace{\gamma_{h\ell}}_{\text{away from HQ}} \underbrace{d_{\ell n}}_{\text{shipping from } \ell} \underbrace{(1 + t_{\ell nk})}_{\text{tariff}} \quad (1)$$

Labour and inputs are bought where the factory sits; shipping is counted from the factory, so one plant serves many countries. Being a multinational shows up in φ_a : head-office services are a non-rival *capability* (complexity α_k penalises stand-alone locals and rewards group capability), never a foreign wage inside a domestic cost — the choice that decides whether the model has one solution, and improves the fit besides.

From: $\gamma_{h\ell}$: Ramondo–Rodríguez-Clare (2013), measured by Head–Mayer (2019). Shipping from the factory: Tintelnot (2017). Inputs: Caliendo–Parro (2015). α_k : Antràs (2003), Antràs–Helpman (2004), used on the capability side.

Entry. Serving market n costs $F_{a,n} = f_k w_\ell$ per period, paid in the same labour as production — so entry costs are real resources, they appear in labour demand, and distributed profits are net of them. Which factories exist is an outcome, not a list.

From: Tintelnot (2017).

Ownership and income. With θ_{gn} the share of group g owned by country n ,

$$X_n = \underbrace{w_n L_n}_{\text{wages}} + \underbrace{\sum_g \theta_{gn} (\Pi_g - \text{entry costs}_g)}_{\text{profits owned, net}} + \underbrace{T_n}_{\text{tariffs}} \quad (2)$$

Profits are earned where factories sell but *received* where owners live; and θ varies firm by firm rather than being one number per country.

From: The firm-level θ is this project’s own ingredient — matched Orbis/D&B–customs records for ten Latin American origins; nothing in the prior literature carries it.

Timing. The model is static. Within the period: capability draws are realised; given wages, prices and market sizes, each group chooses which factories operate in which markets; the operating factories compete in every market; and wages, incomes and price indices adjust until all markets clear. Nothing is sequential in equilibrium.

2.2 Markets: Markups, Profits, and Shares

The mechanism and the markup. A group expanding one factory also steals sales from its *own* other factories, and it counts that loss. So the bigger the group, the more it holds back:

$$\underbrace{\frac{1}{\mu(S)} = 1 - \frac{1-S}{\sigma} - \frac{S}{\eta}}_{\text{quantity competition}} \quad \underbrace{\mu(S) = \frac{\sigma - (\sigma - \eta)S}{\sigma - (\sigma - \eta)S - 1}}_{\text{price competition}} \quad (3)$$

where S_g is the group's share of market sales. Both forms rise from $\sigma/(\sigma - 1)$ at $S = 0$ toward $\eta/(\eta - 1)$ at $S = 1$; every result below holds for *any* increasing μ , so how firms compete is a parameter, not a commitment.

From: Atkeson and Burstein (2008), with the markup moved from the factory to the *owner* — the internalisation Yang (2023) calls cannibalisation.

Profit. With margin $L(S) \equiv 1 - 1/\mu(S)$ and market spending E :

$$\Pi_g = E S_g L(S_g) = \frac{E}{\sigma} S_g (1 + \kappa S_g) \text{ under quantities,} \quad \kappa \equiv \frac{\sigma}{\eta} - 1. \quad (4)$$

Profit rises *faster* than size; the S^2 term is the entire content of the paper. Conduct only scales it ($\kappa = 4.0$ quantities vs 0.8 prices at $\sigma = 5, \eta = 1$).

One market, one unknown. With $K_g \equiv \sum_{i \in g} c_i^{1-\sigma}$ (the group's *strength*), $A \equiv \sum_j p_j^{1-\sigma}$ (how *tough* the market is), and $\psi(S) \equiv S \mu(S)^{\sigma-1}$:

$$\boxed{S_g = \psi^{-1}(K_g/A), \quad \sum_g S_g = 1.} \quad (5)$$

However many factories a group owns, its whole position is one number, K_g — cannibalisation is already inside it — and all it needs to know about rivals is A . A market with hundreds of factories is a problem in one unknown; that is what makes the theorems possible.

2.3 General Equilibrium

An equilibrium is a set of wages, price indices, market sizes, incomes, an active set of factories, and quantities such that every group best-responds in every market, shares sum to one, no group wants to open or close a factory, price indices are consistent with costs, every worker is employed, and every country's budget balances. One labour-market condition is redundant by Walras' law, so one wage is the numeraire. All accounting identities hold to 10^{-15} in the program, at wages far from equilibrium.

3 The Measurement Result

For a country H owning shares θ_g in a market, with $\bar{\theta} = \sum_g S_g \theta_g$ and $\text{Cov}(\theta, S) = \sum_g \theta_g S_g^2 - \bar{\theta} \text{HHI}$:

Proposition 1 (The correct adjustment is a concentration measure).

$$\frac{\Pi_H}{E} = \frac{1}{\sigma} \left[\bar{\theta} + \kappa (\bar{\theta} \text{HHI} + \text{Cov}(\theta, S)) \right], \quad \boxed{\text{error of the country-level calculation} = \frac{E\kappa}{\sigma} \text{Cov}(\theta, S).} \quad (6)$$

Proof. Insert $L = (1 + \kappa S)/\sigma$ into (4), sum with weights θ_g , and subtract $\bar{\theta}$ times total profit $\Pi/E = (1 + \kappa \text{HHI})/\sigma$. $\square$

What matters is not what fraction of *firms* a country owns but what fraction of the *big* ones. With constant markups ($\kappa = 0$) the correction vanishes — which is why the standard workhorse models cannot host the question. **Measured in our data** (market = destination×product×year, group = ultimate parent): the country-level calculation understates the United States' stake by 10%, overstates Argentina's and Peru's by 24–30% — the sign varies owner by owner, which no country-level share can express.

4 Existence and Uniqueness

Theorem 1 (One market). *Given the operating factories (≥ 2 groups), the market equilibrium exists and is unique: ψ is increasing, so $\sum_g \psi^{-1}(K_g/A)$ falls strictly in A and crosses 1 exactly once; group profit is strictly concave in the group’s own quantities.*

Theorem 2 (Entry). *Write the entry payoff as $\Omega(K; A) = \Pi(\psi^{-1}(K/A))$. Under quantity competition with $\eta = 1$, with $a = 2(\sigma - 1)$, $b = \sigma - 2$:*

$$(A) \ \Omega_{KK} < 0: \quad \varepsilon_G(S) = S \left[\frac{a}{1+aS} - \frac{b}{1+bS} - \frac{\sigma}{1-S} \right] < 0 \text{ at every } S \ (\sigma \geq 2);$$

$$(B) \ \Omega_{KA} < 0: \quad \varepsilon_G + \varepsilon_\psi = \frac{aS}{1+aS} - \frac{bS}{1+bS} + \frac{1-2S}{1-S} > 0 \text{ at every } S \leq \frac{1}{2}, \text{ with slack exactly } 1/\sigma \text{ at } S = \frac{1}{2}.$$

If no group’s share exceeds one half on the region between candidate outcomes (the computed solution sits at ≈ 0.30 against an exact frontier of 0.545), there is at most one entry outcome — with no order of moves and no selection rule, and with groups still competing with themselves.

Proof idea. (A) makes each group’s choice a cutoff on its own ladder of factory sets; (B) makes that cutoff fall when the market toughens; and A rises when any group strengthens — so two equilibria contradict each other. Brute force over *every* configuration: 120 of 120 random markets have exactly one outcome.

From: The standard fixes fail here: cost-ranked entry (Gaubert–Itskhoki 2021) needs every entrant separate, and an assumed order of moves (Yang 2023) makes market structure an artifact of the order. The theorem replaces both.

Wages, the last layer. Input prices are a contraction (modulus $\max_k \nu_k < 1$); incomes are a linear system. For wages, let $M_{nj} = \partial \ln W_n / \partial \ln w_j$ (unit row sums by homogeneity). The solver’s own map $\Psi_\kappa(w)_n = w_n (\text{LD}_n / L_n)^\kappa$ has log-Jacobian $(1 - \kappa)I + \kappa M$; wherever that matrix is entrywise positive — gross substitutes off the diagonal, enough damping on it ($\kappa < 1/(1 - \min_n M_{nn})$, a bound the model itself supplies) — Birkhoff’s (1957) theorem makes Ψ_κ a strict contraction in Hilbert’s metric: **at most one equilibrium there, with a geometric rate.** The positivity holds under two checked conditions (no group above half a market — the same one half as Theorem 2 — plus a demand condition), holds at every point tested out to wage spreads of e^2 , and is **computer-certified on the continuum** of every wage vector within $\pm 15\%$ of the solution by outward-rounded interval arithmetic (Moore 1966) — no gap between grid points. What is *not* available is an unconditional theorem, for a stated reason: the one channel that can break the condition is the ordinary income effect, which is this paper’s own subject. A counterexample and the full honesty ledger are in the guide. Ten dispersed starts reach the same wages *and the same set of operating firms*.

5 Quantification: The Six Facts

#	Fact	Model	Data	Status
1	MNE export share; foreign dominant	0.69 (0.54/0.15)	0.46–0.74	level fitted; split produced
2	foreign firms in complex goods	+0.42	+0.43	fitted
3	parents from few countries	0.27	0.13	produced , overshoots
4	grouping by owner raises concentration	$\times 1.94$	$\times 1.12$	produced — a prediction
5	MNEs present $\Rightarrow$ local exports	+46% [†]	positive	positive in the λ config.
6	distance matters less for MNEs	−0.52/+0.70	−0.16/+0.05	produced , too large

Three parameters are fitted (entry cost, head-office disadvantage, pool scaling); Facts 3, 4 and 6 are genuine predictions.

Fact 5, honestly. The empirical fact survives a saturated destination $\times$ product $\times$ year re-estimation at half its published size, so it is real — and the *core* model gets it backwards (−97%): with market spending fixed, entry is zero-sum, and no cost-side spillover can fix that (measured, not assumed). What fixes it is the missing margin: a **rest-of-world supplier in every market**, so that sample exporters hold the small in-sample share $\lambda \approx 0.07$ seen in the data and business stealing falls on the outside supply. [†]With that supplier plus two spillovers to single-plant locals (productivity: Javorcik 2004; export infrastructure: Aitken–Hanson–Harrison 1997), Fact 5 turns **positive on both margins** (+46%; Fact 1 = 0.49 with the split intact, Fact 4 improves to $\times 1.67$, one sign inside Fact 6 flips — disclosed). This *six-fact configuration* (`julia mne_model.jl lambda`) costs three more fitted parameters and is reported beside the core, not in its place: six facts against six fitted parameters is a different claim from five against three, and the reader should see both.

Everything here is one file, base Julia, no packages (`mne_model.jl`); every number above has a committed run behind it. The full paper — every proof written out, every choice justified where it is made, and a provenance table naming the source of each component — is `model_guide.pdf`.